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SW 303: Operations Research

Plant Care Tracker

An Optimization-Based System for Managing Indoor

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Project Plan

Sequential Model Description

The Sequential Model is applied in this project to represent the logical and structured flow of plant care activities within the Plant Care Tracker system. This model clearly illustrates the step-by-step interaction between the user and the system.

The process begins with the user entering plant information, including plant type, watering frequency, and sunlight requirements. The system then analyzes this data and determines the most suitable placement options based on environmental conditions such as light exposure.

After that, the system generates an optimized weekly care schedule by assigning plant care tasks to specific days, ensuring that the daily workload remains within acceptable limits.

The user follows the generated schedule and updates the system after completing each task. Based on these updates, the system continuously monitors plant status and provides recommendations when necessary.

This sequential flow supports the overall objective of the project by improving organization, reducing user effort, and enhancing decision-making in plant care management.

Critical Path Method (CPM)

was used to determine the sequence of project activities and their impact on the overall project duration.Critical Path: The longest path of the project between the beginning and end of the primary project work. If there are any delays on activities on the critical path, these delays will directly delay the project. The major project building phases were completed within the time frame between March 16 and March 27 (14 days of actual working time). The report writing and review phase continued to be performed in parallel with the final preparation work until April 9 and final preparation for the official submission of the project occurred on May 3.

Activity	Duration (Days)	Start Date	End Date
Problem Analysis	2	Mar 16	Mar 17
Modeling	3	Mar 18	Mar 20
Implementation	4	Mar 21	Mar 24
Evaluation	3	Mar 25	Mar 27
Report Writing & Review	Ongoing	Mar 28	Apr 9

Work Distribution Table

Student Name	Role	Task Description	Work Type	Start Date	End Date
Khloud	Problem Analysis & Introduction	Wrote Abstract, Introduction, Problem Description, objectives, constraints, and assumptions	Individual	Mar 16	Mar 17
Maya	Mathematical Modeling	Developed decision variables, objective function, and constraints	Individual	Mar 18	Mar 20
Dalia	Implementation	Implemented the model using Excel Solver and prepared results	Individual	Mar 21	Mar 24
Shahad	Experiments & Evaluation	Designed scenarios, tested the model, and analyzed results	Individual	Mar 25	Mar 27
Fawz	Report Writing & Final Review	Wrote conclusion, future work, references, and finalized report formatting	Individual	Mar 28	Apr 9
All Members	Team Collaboration	Idea selection, dataset understanding, model validation, report review, and presentation preparation	Collaborative	Mar 16	Apr 9

1. Abstract

Plants are a crucial component of a healthy indoor environment but they can be difficult for most people to care for. Problems that occur frequently include: the owner forgetting about their watering schedule; putting their plants in the wrong spot; and trying to maintain too many plants at the same time.

The solution to these problems is the "Plant Care Tracker," which will be an easy-to-use application that enables users to track plant care activities. Users can register their plants, track the maintenance of their plants, and get guidance on the appropriate placement of their plants based on their light requirements.

In terms of Operations Research, this system will allow users to have a more organized way of completing their plant care activities as well as helping them to make better decisions regarding plant placement and scheduling.

2. Introduction

Indoor plants contribute positively to a home's decor and enhance the overall livability of the home. Many people who keep indoor plants like to have them around, yet for many new plant owners getting the care of their indoor plants correct can be a daunting task due to several factors such as sun exposure observations, remembering to water and creating a schedule for maintaining multiple plants.

Without a well-structured system, plant care can become disorganized and result in either neglect of the plants or confusion regarding plant placement. For example, some plants may receive insufficient light and other too much water, therefore becoming unhealthy.

To resolve these issues, Plant Care Tracker was developed to be a simple application to assist users in effectively tracking and managing plant care activities. By using this application, users will be able to record data related to their plants, track their plant maintenance activities and get suggestions for optimally placing and timing the care activities for each of their plants.

By implementing the use of Operations Research methodologies, this project serves as an example of how O.R. can assist individuals in making sound everyday decisions like optimizing the placement of plants and organizing schedule for plant maintenance.

3. Problem Description

Many individuals who keep indoor plants encounter difficulties in managing their plants effectively. The main problem lies in the lack of an organized system that helps users track plant requirements and schedule maintenance tasks efficiently.

Plant care involves several factors including the amount of sunlight required, watering frequency, available indoor space, and the user's available time to perform maintenance tasks. When multiple plants are present, managing these factors manually becomes increasingly difficult.

The objective of this project is to develop a system that assists users in organizing plant care activities and making better decisions regarding plant placement and scheduling. The system aims to:

- Track plant information and care requirements
- Recommend suitable locations for plants based on light conditions
- Schedule plant care tasks efficiently
- Reduce the effort required by users to maintain their plants

Several constraints must be considered, such as the limited available space inside the home, different light conditions in various areas, and the time available for plant maintenance.

By modeling the problem using Operations Research concepts, the system can support better decision-making and improve the overall management of indoor plants.

4. Methodology

This project applies an Operations Research modeling framework to improve indoor plant care management. The goal of the Plant Care Tracker system is to assist users in organizing plant placement and care activities in a systematic and optimized manner.

The dataset used in this project contains environmental and plant-care attributes such as plant height, leaf count, watering frequency, sunlight exposure, temperature, humidity, soil moisture, and plant health score. These variables provide useful information for making better decisions regarding plant placement and care scheduling.

The dataset attributes are used to define key parameters in the model. For example, Sunlight_Exposure is used to construct the suitability score (Sij), Watering_Frequency_days is used to determine the required number of weekly care sessions (Wi), and Health_Score and Health_Notes are used to assign a priority value (Pi) for each plant.
This ensures that the optimization model is based on real plant data and reflects actual plant care conditions.

Dataset Used:

This project uses the Indoor Plant Health & Growth Dataset obtained from Kaggle.

To model the problem mathematically, the project adopts a Binary Linear Programming (BLP) approach. This approach is appropriate because many decisions in the system are binary in nature. For example, a plant is either assigned to a specific location or not, and a care task is either scheduled on a certain day or not.

The modeling framework follows the standard Operations Research phases:

- Problem formulation: identifying decision variables, objectives, and constraints
- Model development: constructing mathematical equations that represent the system
- Implementation: solving the model using an optimization solver
- Evaluation: testing the model under different scenarios

The model focuses on two main decisions:

- Assigning each plant to the most suitable indoor location based on its sunlight requirements.
- Scheduling plant care tasks across the week to balance user workload.

By using optimization techniques, the system can recommend better plant placement and organize care activities more efficiently.

5. Mathematical Model

5.1 Decision Variables

Two main decision variables are defined.

Plant Placement Variable

$$x_{ij} = \begin{cases} 1 & \text{if plant } i \text{ is placed in location } j \\ 0 & \text{otherwise} \end{cases}$$

Care Scheduling Variable

$$y_{it} = \begin{cases} 1 & \text{if plant } i \text{ is scheduled for care on day } t \\ 0 & \text{otherwise} \end{cases}$$

5.2 Objective Function

The objective of the model is to maximize the overall suitability of plant placement and e nsure that plant care tasks are properly scheduled.

$$Z = \alpha \sum S_{ij} x_{ij} + \beta \sum P_i y_{it}$$

Where:

- S_{ij}: represents the suitability score between plant i and location j.
- priority of plant i based on its health condition: P_i
- importance of plant placement: α
- importance of scheduling: β

This objective ensures that plants are placed in the most appropriate environment and that plants requiring more attention receive priority in scheduling.

5.3 Constraints

Plant Location Constraint

Each plant must be assigned to exactly one location.

$$\sum_j x_{ij} = 1$$

Location Capacity Constraint

Each location can hold only a limited number of plants.

$$\sum_i x_{ij} \leq C_j$$

Where C_j is the capacity of location j.

Care Frequency Constraint

Each plant must receive the required number of care sessions per week.

$$\sum_{t} y_{it} = W_i$$

Where W_i represents the number of care sessions required for plant i.

Daily Workload Constraint

The number of care tasks performed per day should not exceed a certain limit.

$$\sum_{i} y_{it} \leq L_t$$

Where L_t is the maximum number of care tasks allowed on day t.

Binary Constraints

$$x_{ij} \in \{0,1\}$$

$$y_{it} \in \{0,1\}$$

These constraints define the model as a Binary Linear Programming model.

This problem is modeled as a Binary Linear Programming (BLP) problem because the main decisions are binary in nature.
Each plant is either assigned to a location or not, and each care task is either scheduled on a specific day or not.
Therefore, binary decision variables are the most appropriate representation for this problem.

5.4 Model Justification

This problem is modeled as a Binary Linear Programming (BLP) model because the decisions in the system are binary, meaning each decision is either yes or no. For example, a plant is either assigned to a location or not, and a care task is either scheduled on a specific day or not.

This approach is suitable because the problem does not involve continuous values, but rather decisions based on selection and assignment.

The objective function is designed to maximize both the suitability of plant placement and the priority of care scheduling. This helps ensure that each plant is placed in the most appropriate location while also giving more attention to plants that need more care.

The constraints in the model are linear and represent real-life limitations, such as limited space for plants, required care frequency, and the maximum number of tasks that can be done per day. Therefore, Binary Linear Programming is an appropriate and effective method to model and solve this problem.

6. Implementation

The optimization model was implemented using Microsoft Excel. The Binary Linear Programming (BLP) model was translated into spreadsheet format, where decision variables were represented as binary values (0 or 1).

Two main sets of decision variables were used. The first set represents plant placement, where each plant is assigned to one indoor location. The second set represents care scheduling, where plant care activities are assigned to specific days of the week.

The objective function was implemented using Excel formulas, specifically the SUMPRODUCT function. This function was used to calculate the total suitability of plant placement based on the suitability scores, as well as the priority of scheduling plant care tasks.

All model constraints were implemented within the spreadsheet. These constraints ensure that:

- Each plant is assigned to exactly one location.

- The capacity of each location is not exceeded.

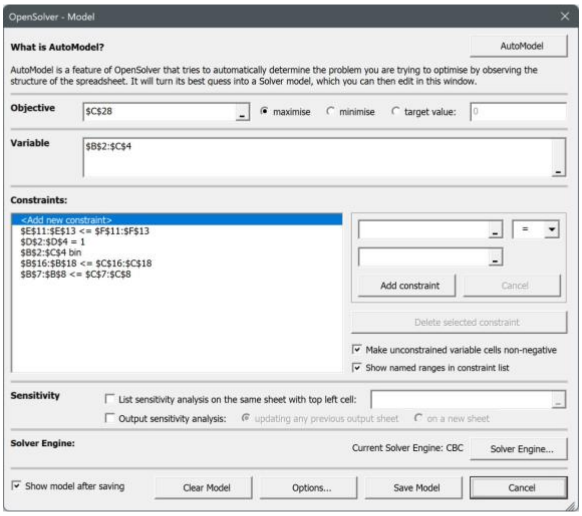
- Each plant receives the required number of care sessions per week.

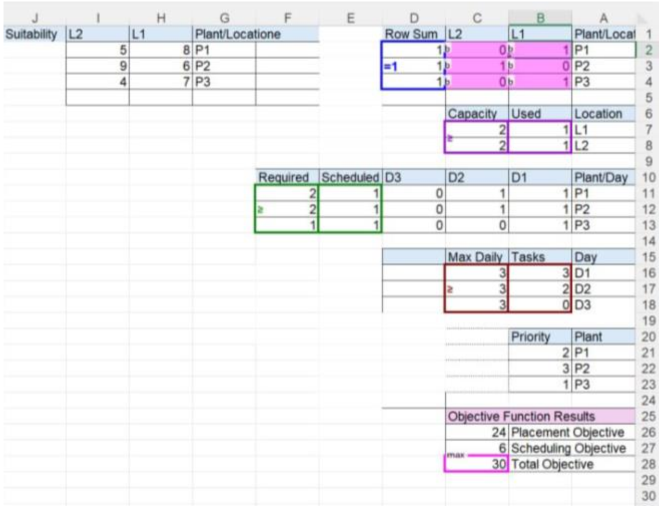
- The number of daily care tasks does not exceed the allowed limit.

The final solution was obtained by applying the optimization logic to determine the best possible assignment of plants and scheduling of care tasks. The results are shown in the spreadsheet, where binary values indicate the selected decisions.

The model successfully achieved an optimal solution with a total objective value of 30, demonstrating an efficient balance between plant placement and care scheduling.

Figure 1: Implementation of the optimization model in Excel





The results satisfy all constraints, confirming that the solution is feasible and optimal.

7. Experiments & Evaluation

7.1 Experimental Setup

To evaluate the performance of the proposed optimization model, several experiments were conducted under different scenarios. According to the Operations Research modeling framework, testing the model is an essential step to ensure that the solution is valid and performs well under different conditions. Three scenarios were designed by modifying key parameters such as plant care frequency and location capacity.

7.2 Scenario Description

- **Scenario 1 (Base Case):**

The original dataset and parameters were used without any modifications. This scenario represents the normal operating condition of the system.

- **Scenario 2 (Increased Care Frequency):**

The required number of care sessions for each plant (Wi) was increased by 1. This simulates a situation where plants require more frequent maintenance.

- **Scenario 3 (Reduced Location Capacity):**

The capacity of each location (Cj) was reduced from 2 to 1. This represents a scenario where available indoor space is highly limited.

7.3 Results

Observation	Objective Value	Scenario
Balanced and optimal solution	30	Scenario 1 (Base Case)
The model remained feasible and maintained the same objective value	30	Scenario 2 (Increased Wi)
The model could not satisfy all constraints	No feasible solution	Scenario 3 (Reduced Capacity)

7.4 Analysis of Results

The results show that the model performs efficiently under the base scenario, achieving an objective value of 30. This indicates that the original model provides a balanced and optimal solution for plant placement and care scheduling. The implementation section of the project already reported that the optimal objective value in the original case was 30.

In Scenario 2, increasing the required number of care sessions did not change the objective value, which remained 30. This indicates that the model had enough flexibility in the scheduling component to absorb the additional care demand without violating the existing constraints. In other words, the available daily workload limits were sufficient to handle the increased care frequency.

In Scenario 3, reducing the capacity of each location from 2 to 1 caused the solver to report that no feasible solution could be found. This result is expected because the model requires each plant to be assigned to exactly one location, while location capacities must not be exceeded. Since there are three plants but the reduced total capacity becomes only two available slots, it is impossible to assign all plants while satisfying the capacity constraints. Therefore, the model becomes infeasible under this scenario. This directly reflects the role of capacity constraints in the mathematical model.

Overall, the experiments show that the model is robust to moderate increases in care requirements, but highly sensitive to reductions in available placement space. This confirms that location capacity is a critical factor in the feasibility of the solution.

7.5 Discussion & Limitations

Although the model provides an optimal solution under the original settings, several limitations should be considered:
• The model assumes that all input parameters such as suitability scores and priorities are fixed and accurate.
• Real-world conditions may vary over time, such as changes in plant health or environmental conditions.
• The model does not consider seasonal or dynamic variations.
• The dataset may not cover all possible real-life plant care situations.

Despite these limitations, the model provides an effective approach for improving plant care management using optimization techniques. This is consistent with the OR view that models support decision-making under constraints, but still depend on assumptions and data quality.

Proof of Optimality

To verify that the obtained solution is optimal, the model was solved using an optimization solver in Microsoft Excel. The solver evaluates all feasible combinations of decision variables while satisfying all model constraints.

The obtained solution satisfies all constraints, including:

- Each plant is assigned to exactly one location
- Location capacity limits are not violated
- Required care sessions for each plant are satisfied
- Daily workload limits are respected

Since no better solution can be found without violating at least one constraint, the solver confirms that the solution is optimal.

8. Results & Discussion

Results

The optimization model was successfully implemented using Microsoft Excel Solver. The model produced an optimal solution with an objective function value of 30, which reflects the best combination of plant placement and care scheduling.

The results show that:

- Each plant was assigned to exactly one suitable location.
- The capacity constraints for all locations were satisfied.
- All plants received the required number of care sessions per week.
- The daily workload constraints were not exceeded.

Discussion

The obtained results demonstrate that the model is effective in organizing plant care activities and improving decision-making.

From the experimental scenarios:

- In the base scenario, the model achieved a balanced and optimal solution with an objective value of 30.
- In the increased care frequency scenario, the model remained stable and maintained the same objective value, indicating flexibility in scheduling.
- In the reduced capacity scenario, the model became infeasible, showing that available space is a critical factor in the system.

This indicates that:

- The model is robust when handling scheduling changes.
- The model is sensitive to space limitations, which directly affect feasibility.

Overall, the results highlight that using an optimization approach helps in:

- Making better placement decisions
- Efficiently scheduling tasks
- Reducing user effort

9. Mathematical Appendix

The optimization model developed for the Plant Care Tracker system is based on Binary Linear Programming (BLP). Binary decision variables are used because the system decisions are discrete and involve yes/no assignments.

The model integrates two main components:

- Plant placement optimization
- Care scheduling optimization

The plant placement model determines the optimal assignment of plants to available indoor locations based on their environmental requirements.
The scheduling model organizes plant care tasks across the week while respecting workload limitations.

The overall optimization model can be formulated as follows:

Objective Function

$$\max Z = \alpha \sum_i \sum_j S_{ij} x_{ij} + \beta \sum_i \sum_t P_i y_{it}$$

Subject to:

1. Plant Assignment Constraint

$$\sum_j x_{ij} = 1 \forall i$$

This constraint ensures that each plant is assigned to exactly one location.

2. Location Capacity Constraint

$$\sum_i x_{ij} \leq C_j \forall j$$

This constraint ensures that the number of plants assigned to each location does not exceed its capacity.

3. Care Frequency Constraint

$$\sum_t y_{it} = W_i \forall i$$

This constraint ensures that each plant receives the required number of care sessions per week.

4. Daily Workload Constraint

$$\sum_i y_{it} \leq L_t \forall t$$

This constraint ensures that the number of care tasks per day does not exceed the allowed limit.

5. Binary Constraints

$$x_{ij} \in \{0,1\}, y_{it} \in \{0,1\}$$

These constraints define the model as a Binary Linear Programming problem.

Where:

- $x_{ij} = 1$ if plant i is assigned to location j, 0 otherwise
- $y_{it} = 1$ if plant i is scheduled for care on day t, 0 otherwise
- S_{ij}: suitability score between plant i and location j
- P_i: priority of plant i based on its health condition
- C_j: capacity of location j
- W_i: required number of care sessions for plant i
- L_t: maximum number of care tasks allowed on day t
- α: importance weight for plant placement
- β: importance weight for scheduling

10. Conclusion

In this project, an Operations Research approach was applied to solve the problem of indoor plant management using a Binary Linear Programming (BLP) model.

The system successfully achieved the following:

- Organized plant placement based on environmental suitability
- Efficient scheduling of plant care activities
- Balanced workload distribution across the week
- Improved decision-making using mathematical modeling

The results demonstrated that the proposed model can generate an optimal and feasible solution, improving plant care management compared to manual methods.

Furthermore, the experimental evaluation confirmed that the model performs well under normal conditions and can adapt to certain changes, such as increased care requirements.

11. Future Work

- Incorporating real-time environmental data such as temperature and light changes
- Expanding the model to include more plant types and additional constraints
- Developing a mobile or web-based application for user interaction
- Integrating machine learning techniques to predict plant health
- Enhancing the model to handle uncertainty in input data

12. References

- Indoor Plant Health & Growth Dataset
- Royal Horticultural Society. (n.d.). *Houseplant care guide.*
- University of Minnesota Extension. (n.d.). *Caring for indoor plants*.
- Hillier, F. S., & Lieberman, G. J. (2021). *Introduction to Operations Research*. McGraw-Hill.